Bulk RNA-seq Differential Expression and Pathway Analysis Report

Bioinformatics Analysis Service

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1. Executive Summary

This report presents a comprehensive bulk RNA-seq analysis of MCF-7 breast cancer cells cultured under control conditions and in coculture with adipose-derived stem cells (ADSCs).

The analysis identifies transcriptional changes associated with tumor microenvironment interactions, including differential gene expression, functional enrichment, and pathway-level reprogramming.

Key findings indicate strong modulation of extracellular matrix organization, hypoxia response, metabolic reprogramming, and pro-growth signaling pathways.

2. Experimental Design

The dataset consists of 6 RNA-seq samples:

Condition	Replicates	Description
Control	3	MCF-7 cells cultured alone
ADSC	3	MCF-7 cells cocultured with ADSCs

No batch effects were reported unless otherwise specified.

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3. Data Processing Workflow

RNA-seq preprocessing was performed using a standard pipeline:

- Quality control: FastQC / MultiQC
- Adapter trimming: Trimmomatic
- Alignment: HISAT2
- Gene quantification: featureCounts

Raw counts were used for downstream statistical analysis.

Analysis was performed following established Bioconductor workflows for bulk RNA-seq differential expression analysis, ensuring reproducibility, analytical transparency, and adherence to widely accepted computational standards.

3.1 Statistical framework

Differential expression analysis was performed using edgeR, which models count data using a negative binomial distribution. Library normalization was performed using the TMM method to account for compositional differences between samples.

Statistical significance was defined using the false discovery rate (FDR) framework to control for multiple hypothesis testing.

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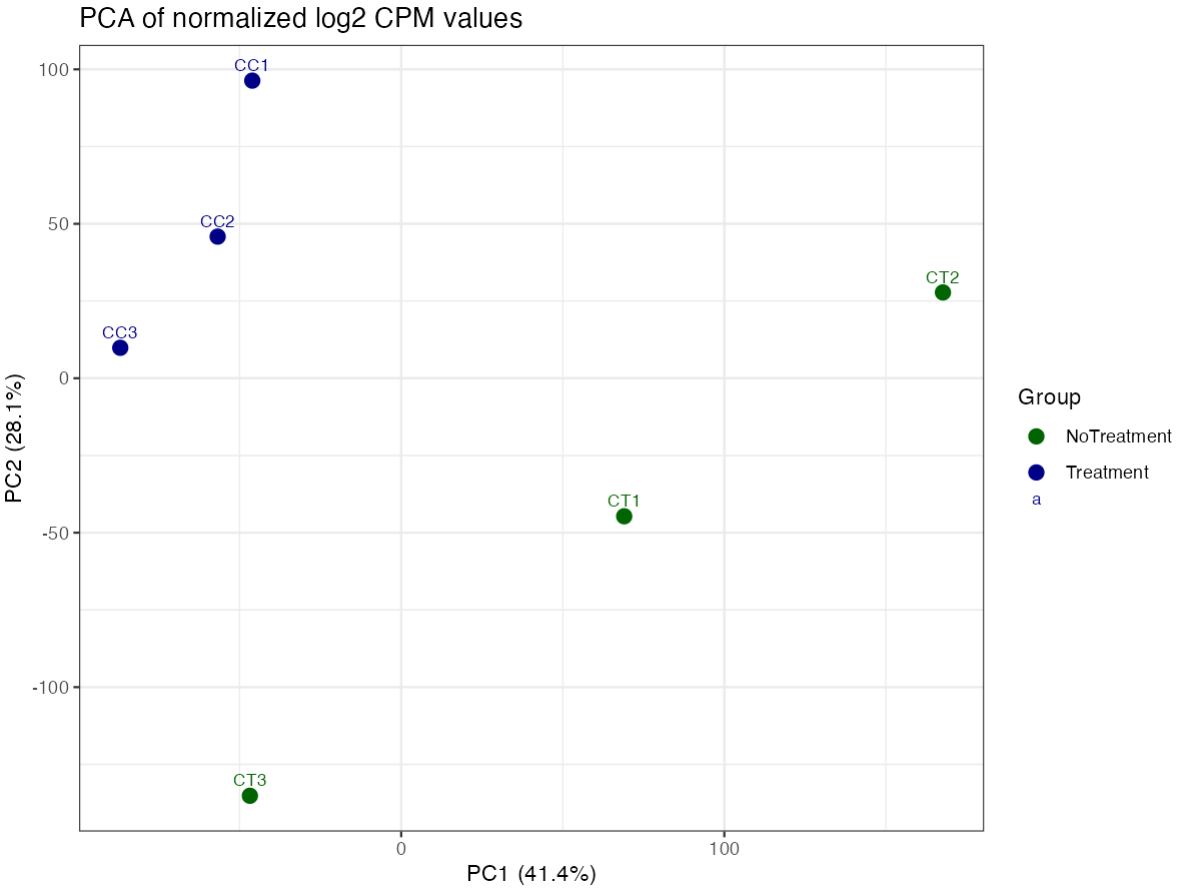
4. Exploratory Data Analysis (EDA)

4.1 Principal Component Analysis (PCA)

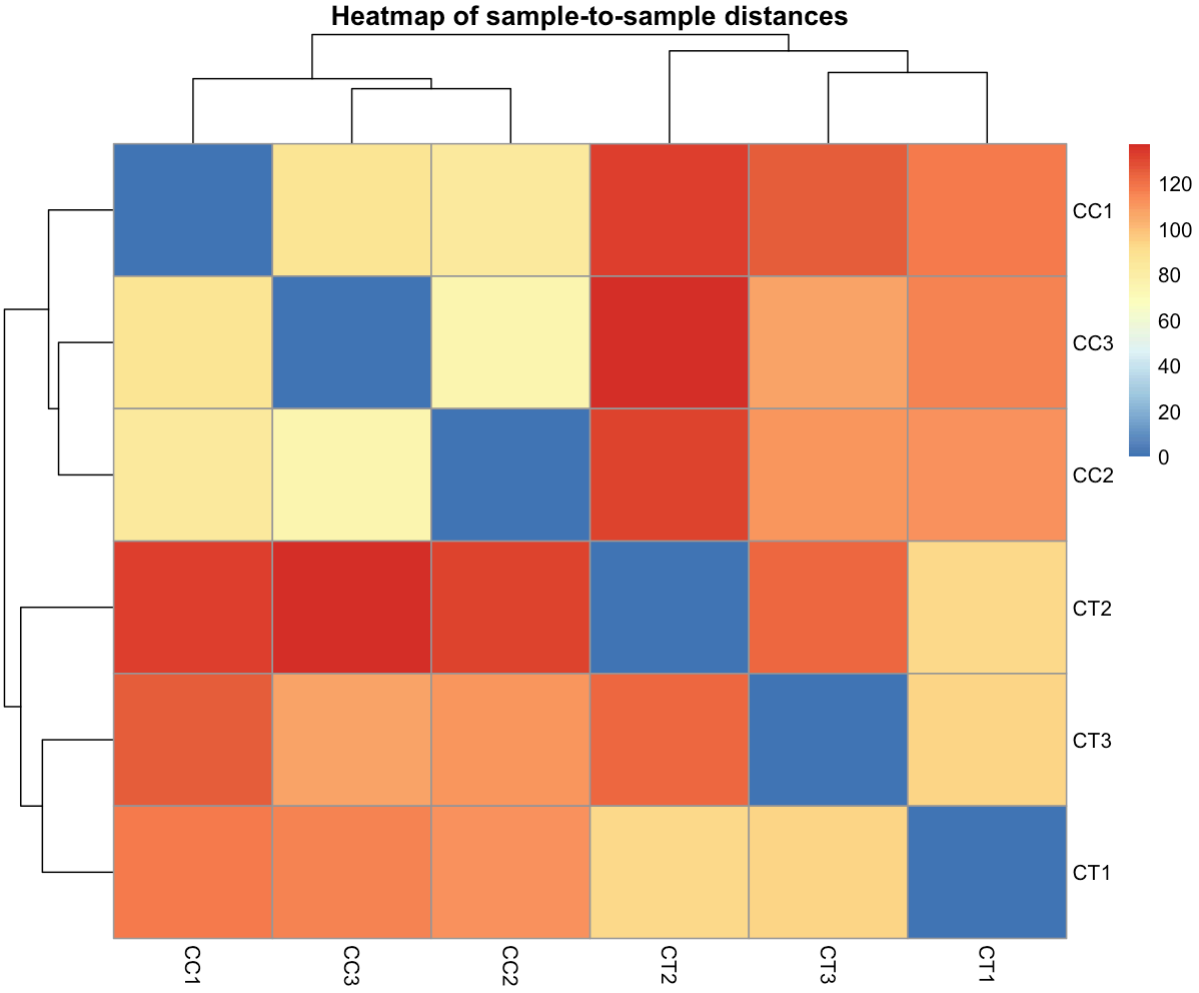
PCA was used to assess global transcriptional differences between conditions.



PCA revealed a clear separation between control and ADSC-treated samples, indicating strong global transcriptional reprogramming induced by tumor microenvironment interactions.

4.2 Sample Clustering

Sample-distance heatmap was performed to assess sample similarity.



5. Differential Expression Analysis (DEGs)

Differential expression analysis was performed using edgeR with TMM normalization.

Criteria for significance:

```
## - FDR < 0.05
```

```
## - |log2FC| > 1
```

5.1 Summary of DEGs

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## Total DEGs identified: 1013
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## Upregulated genes: 887
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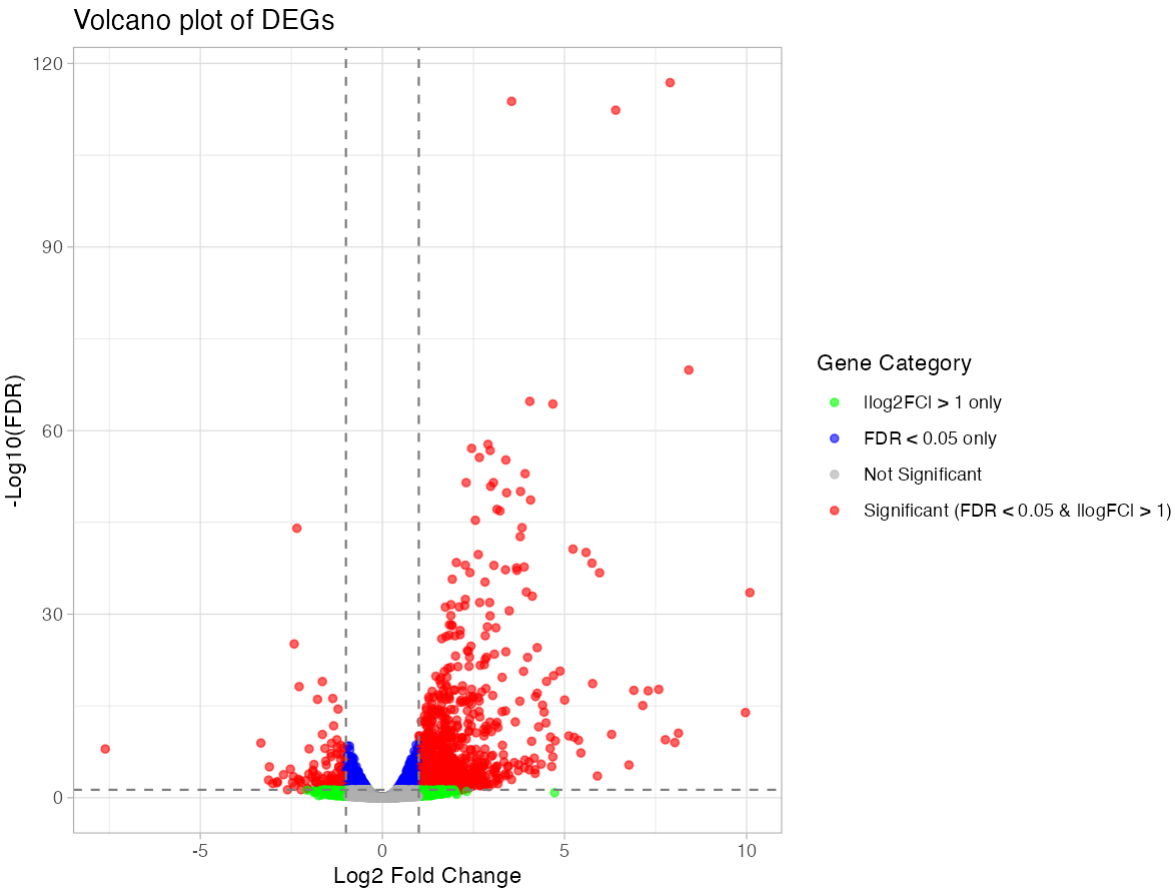
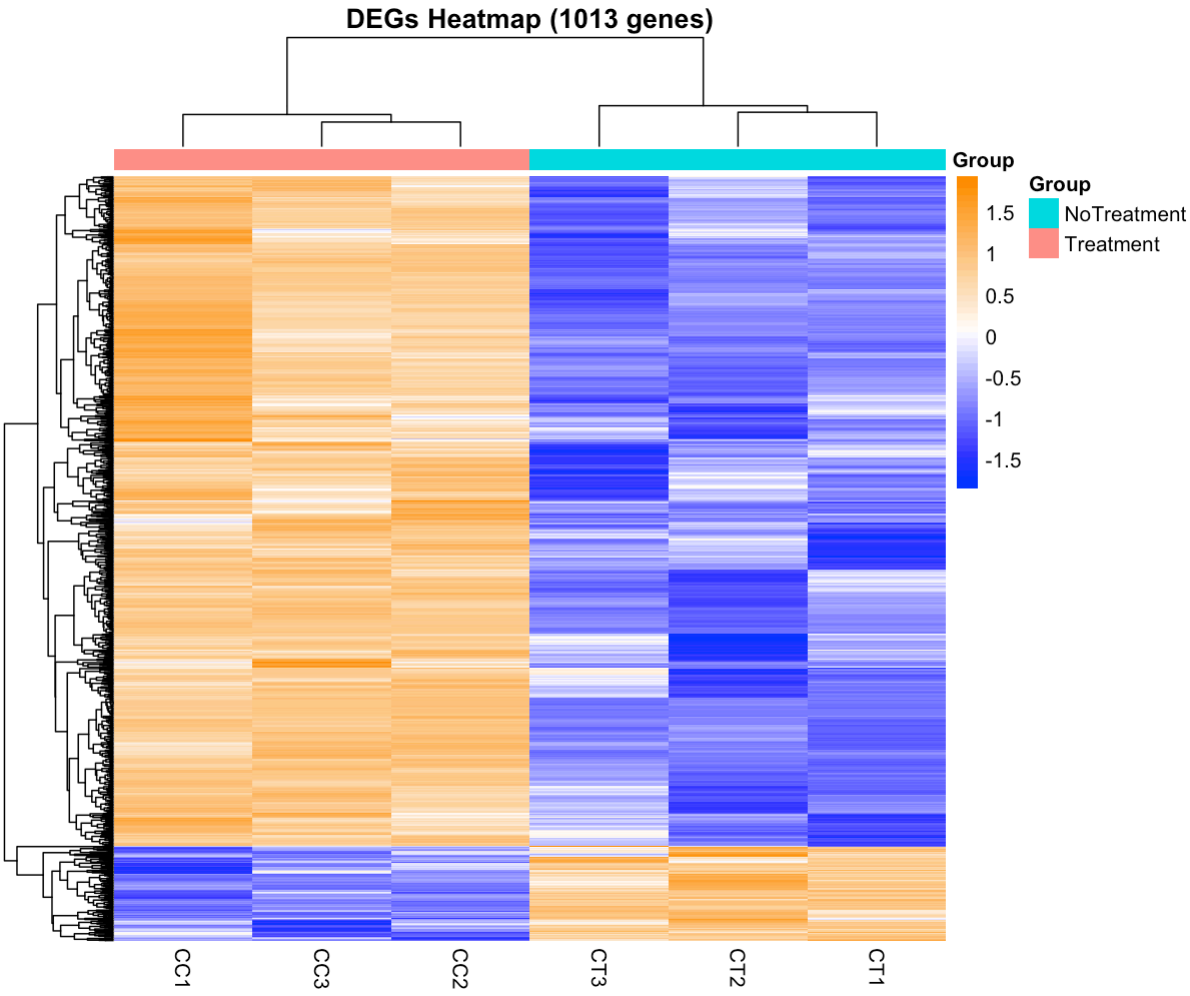
## Downregulated genes: 126
```

5.2 Volcano Plot of DEGs



5.3 Heatmap of DEGs



6. Functional Enrichment Analysis

Functional enrichment analysis was performed using GO, KEGG, Reactome, and GSEA.

6.1 Gene Ontology terms

Biological Process

- Response to hypoxia
- Extracellular matrix organization
- Cellular response to oxygen levels

Molecular Functions

- Extracellular matrix structural constituent
- Monosaccharide binding
- Oxidoreductase activity

Cellular Component

- Vesicle lumen
- Basement membrane
- Extracellular matrix

🧠 Biological interpretation (GO enrichment)

The GO enrichment analysis highlights a coordinated transcriptional response involving hypoxia signaling, extracellular matrix remodeling, and metabolic adaptation. The enrichment of response to hypoxia and cellular response to oxygen levels suggests activation of oxygen-sensing pathways consistent with tumor microenvironment stress conditions.

In parallel, extracellular matrix organization and related structural components indicate active remodeling of the tumor microenvironment, a key feature associated with increased cell adhesion, migration, and stromal interaction in breast cancer progression.

At the molecular level, enrichment of extracellular matrix structural components and oxidoreductase activity reflects extracellular remodeling processes coupled with metabolic and redox adaptation. Finally, the presence of extracellular and vesicle-associated compartments supports enhanced intercellular communication within the tumor microenvironment.

Overall, these results are consistent with early pro-tumorigenic transcriptional reprogramming driven by stromal interaction, as described in ADSC-cocultured MCF-7 models.

6.2 KEGG Pathways

Significant pathways include:

- HIF-1 signaling pathway
- Glycolysis / gluconeogenesis
- Complement and coagulation cascades

6.3 Reactome Pathways

- Interleukin-4 and Interleukin-13 signaling
- Fibronectin matrix formation
- Assembly of collagen fibrils and other multimeric structures

🧠 Biological interpretation (KEGG and Reactome pathway)

The KEGG and Reactome enrichment analyses consistently highlight major biological processes associated with tumor microenvironment remodeling and metabolic adaptation. The enrichment of the HIF-1 signaling pathway together with glycolysis/gluconeogenesis suggests a metabolic shift toward hypoxia-driven glycolytic reprogramming, a well-established feature of early tumor progression and cellular adaptation to low oxygen conditions.

In parallel, pathways such as ECM-receptor interaction, fibronectin matrix formation, and collagen organization indicate strong extracellular matrix remodeling and reorganization of cell–matrix interactions, supporting increased cell adhesion, migration, and stromal communication.

Additionally, the activation of immune-related pathways, including complement and coagulation cascades and interleukin-4/13 signaling, suggests a pro-inflammatory microenvironment potentially driven by stromal–tumor crosstalk.

Overall, these pathways converge on a biological program characterized by hypoxia adaptation, metabolic reprogramming, and extracellular matrix remodeling, consistent with ADSC-induced pro-tumorigenic transcriptional changes in MCF-7 cells.

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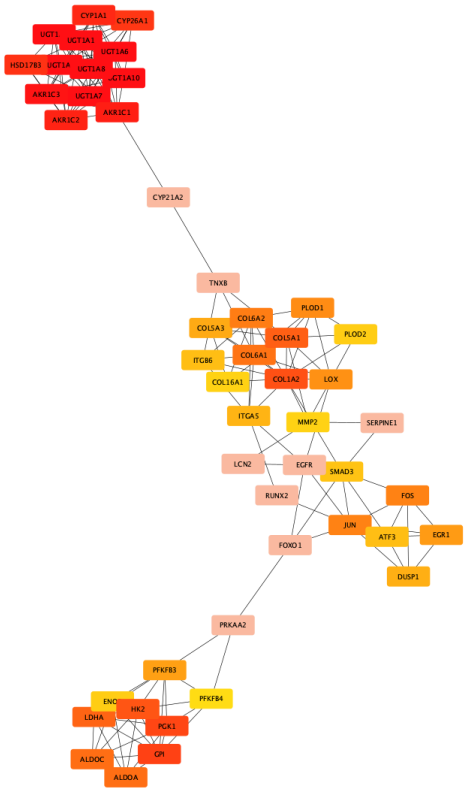
7. Network Analysis

Protein-protein interaction (PPI) analysis was performed using STRING database.

Hub genes were identified using network topology analysis (cytoHubba MCC method)



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8. Integrated Biological Interpretation

The transcriptional profile of ADSC-treated MCF-7 cells reveals coordinated changes across multiple biological systems.

A strong enrichment of extracellular matrix organization and integrin signaling pathways suggests active remodeling of the tumor microenvironment, potentially enhancing cell adhesion and migratory capacity.

In parallel, hypoxia-related and metabolic pathways indicate a shift toward glycolytic metabolism and cellular stress adaptation, consistent with early tumor progression mechanisms.

Additionally, activation of cytokine signaling and JAK-STAT pathways highlights crosstalk between tumor cells and stromal components, suggesting a pro-inflammatory microenvironment.

Finally, enrichment of growth factor signaling and FOXO-mediated transcription suggests enhanced proliferative and survival signaling.

Overall, these results indicate that ADSC coculture promotes a pro-tumorigenic transcriptional program in MCF-7 cells.

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9. Conclusions

1. ADSC coculture induces strong transcriptional reprogramming in MCF-7 cells
2. ECM remodeling and hypoxia response are key biological features
3. Metabolic and inflammatory pathways are significantly altered
4. Results are consistent with a pro-tumorigenic tumor microenvironment effect

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10. Reproducibility

- R version: v.4.5.1
- Main Packages: edgeR, clusterProfiler, ReactomePA, fgsea, WGCNA
- Session information available upon request
- Recommended: use renv for environment reconstruction

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11. Study limitations and considerations

This analysis is based on a limited sample size (n = 3 per group), which is common in RNA-seq experimental designs but may limit statistical power for detecting low-effect genes.

Differential expression analysis was performed using edgeR, which assumes negative binomial distribution and is robust for small sample sizes, but results should be interpreted in the context of biological variability.

No batch effects were detected or reported; however, batch correction methods (e.g., surrogate variable analysis) may be required in more complex experimental designs.

Finally, functional enrichment results depend on existing pathway annotations and should be interpreted as hypothesis-generating rather than definitive biological conclusions

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12. Deliverables (client-ready output)

This analysis provides structured outputs designed for downstream interpretation and decision-making, including:

- Differential expression results with statistical confidence (log2FC, FDR, CPM)
- Ranked gene lists for biomarker prioritization
- Functional pathway summaries highlighting key dysregulated biological processes
- Network-based identification of hub genes potentially involved in regulatory control
- Visual reports (PCA, volcano plots, heatmaps) suitable for publication or presentations
- Biological interpretation summary to support hypothesis generation

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13. Biological and translational relevance

The observed transcriptional changes reflect key features of tumor microenvironment-driven cancer progression, including extracellular matrix remodeling, hypoxia adaptation, and metabolic reprogramming.

These processes are consistent with features associated with increased tumor aggressiveness and may provide insights into early molecular events linked to breast cancer progression in stromal-rich environments.

Rather than isolated pathway effects, the results indicate a coordinated tumor microenvironment-driven transcriptional program organized into three functional biological modules:

1. **Microenvironment remodeling module** (ECM organization, integrin signaling, collagen formation)
2. **Metabolic adaptation module** (HIF-1 signaling, glycolysis, oxidative stress response)
3. **Immune-stromal communication module** (cytokine signaling, complement activation, JAK-STAT pathway)

These findings may provide actionable insights into stromal–tumor interactions and could help prioritize candidate pathways for further experimental validation or therapeutic targeting.

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14. Potential applications

This type of analysis can support:

- Biomarker discovery for cancer progression studies
- Identification of therapeutic targets in tumor microenvironment interactions
- Functional characterization of treatment-induced transcriptional changes
- Preclinical hypothesis generation for breast cancer research
- Integration with proteomics or clinical datasets for multi-omics analysis